Ayushi Singh

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PROFESSIONAL SUMMARY

Recent B.Tech graduate in Computer Science and Artificial Intelligence with hands-on project experience building GenAI applications, AI agents, and RAG-based systems, alongside a strong foundation in Python, NLP, and deep learning. Skilled in data preprocessing, model training, and evaluation, with practical exposure to LLM-powered agent design, vector databases, and text-to-SQL pipelines. Familiar with REST APIs, Git, and AWS basics, and eager to build further expertise in Generative AI and agentic systems.

EDUCATION

Bachelor of Technology, Computer Science and Artificial Intelligence Dec 2022 – May 2026
Pranveer Singh Institute of Technology CGPA: 7.1

PROJECTS

CareCRM AI — AI CRM & Customer Support Copilot
Python, FastAPI, LLM, RAG, Vector DB, CRM APIs

- Built an AI-powered customer support copilot integrating CRM data and knowledge bases to provide context-aware responses to customer queries.
- Implemented RAG, semantic search, and LLM-based response generation to retrieve relevant customer and product information while reducing hallucinated responses.
- Developed tool-calling workflows with FastAPI and CRM REST APIs to retrieve records, summarize tickets, and automate actions, with memory and confidence-based human handoff.

EnterpriseIQ — Enterprise RAG Knowledge Assistant
Python, LangChain, FAISS/ChromaDB, Sentence Transformers, LLM

- Built a document intelligence system for querying enterprise knowledge bases using Retrieval-Augmented Generation (RAG).
- Implemented document ingestion, chunking, embedding generation, semantic retrieval, and reranking to improve context relevance.
- Developed a FastAPI-based backend supporting multi-document search, conversational queries, and source-cited responses, evaluated using relevance and faithfulness metrics to reduce hallucinations.

Guardio — Credit Card Fraud Detection
Python, Scikit-learn, Pandas, NumPy

- Built an end-to-end fraud detection pipeline using Logistic Regression on highly imbalanced transaction data, achieving 97% accuracy and 92% recall on fraudulent cases.
- Applied SMOTE and feature scaling during preprocessing to improve minority-class detection and strengthen rare-event classification performance.

TECHNICAL SKILLS

- **Languages:** Python, JavaScript, SQL, HTML, CSS
- **Generative AI & AI Agents:** LLM Agents, Tool/Function Calling, Text-to-SQL, RAG, Prompt Engineering, LangChain, LlamaIndex, Vector Databases (Pinecone, ChromaDB), Embeddings, LLM Fine-tuning (BERT), Ollama
- **ML / Deep Learning:** Machine Learning, NLP, TensorFlow, PyTorch, Keras, Scikit-learn, Sentence Transformers, Model Evaluation
- **Data & Visualization:** NumPy, Pandas, Matplotlib, Seaborn, Power BI, Tableau, Excel
- **Backend, APIs & Frontend:** FastAPI, REST APIs, OpenAI/Claude API, Postman, React.js, Tailwind CSS
- **Databases:** PostgreSQL, MySQL, MongoDB
- **Developer Tools:** Git, GitHub, VS Code
- **Relevant Coursework:** Data Structures, Object-Oriented Programming, Database Management
- **Soft Skills:** Leadership, Communication, Problem Solving, Decision Making, Innovation

CERTIFICATION

- **HTML5:** Learned HTML5 fundamentals, semantic elements, forms, tables, multimedia, and webpage structure.
- **Python:** Foundational knowledge of Python — variables, conditionals, loops, functions, and basic data structures.
- **SQL:** Demonstrated proficiency in writing SQL queries, filtering data, joins, and basic database operations.
- **Generative AI:** Learned how to design, refine, and optimize prompts for effective interaction with Generative AI models.

EXTRACURRICULAR

- Participated in Startup Charcha, an entrepreneurship-focused session with startup founders and industry professionals.
- Assisted in organizing technical events, seminars, and guest lectures.